

## VIRAC, Philippines

WMO: 984460

Lat: 13.5765N

Lon: 124.2098E

Elev: 121

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |       |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|-------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |       |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR    | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)   | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 71.2       | 72.6 | 66.8                          | 99.2 | 74.1 | 68.2 | 104.0 | 74.9 | 22.6                  | 76.9 | 18.2 | 79.6 | 3.5                   | 320  | 0.337 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 10.4                   | 91.3            | 83.7 | 90.3 | 83.3 | 89.5 | 82.9 | 85.6                | 89.3 | 84.6 | 88.7 | 83.8 | 88.2 | 5.5                  | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 84.7                            | 183.0 | 88.5 | 83.6 | 176.3 | 87.9 | 82.6 | 170.9 | 87.4 | 50.3          | 89.3 | 49.0 | 88.9 | 48.1 | 88.3 | 89.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| 17.1              | 13.5 | 11.8 | 67.8                       | 94.8 | 3.3                | 2.4 | 65.4                                               | 96.5 | 63.5       | 98.0 | 61.7       | 99.3 | 59.3       | 101.1 |    |    |
|                   |      |      | 66.1                       | 86.4 | 2.7                | 1.5 | 64.2                                               | 87.5 | 62.6       | 88.4 | 61.1       | 89.2 | 59.2       | 90.3  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 81.3   | 78.6  | 79.0  | 79.8  | 81.3  | 82.9  | 83.1  | 82.4  | 83.1  | 82.5  | 81.5  | 81.0  | 80.0 |
|                                                                                  | DBStd       | 2.29   | 1.78  | 1.78  | 1.65  | 1.48  | 1.50  | 1.74  | 1.74  | 1.95  | 1.85  | 1.56  | 1.51  | 1.79 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD50       | 11421  | 887   | 811   | 925   | 939   | 1021  | 994   | 1005  | 1027  | 975   | 977   | 931   | 929  |
|                                                                                  | CDD65       | 5946   | 422   | 391   | 460   | 489   | 556   | 544   | 540   | 562   | 525   | 512   | 481   | 464  |
|                                                                                  | CDH74       | 61930  | 3260  | 3195  | 4319  | 5318  | 6678  | 6558  | 6088  | 6510  | 5805  | 5255  | 4757  | 4187 |
|                                                                                  | CDH80       | 19377  | 485   | 633   | 1102  | 1782  | 2615  | 2569  | 2101  | 2441  | 2010  | 1543  | 1251  | 845  |
| Wind                                                                             | WSAvg       | 5.1    | 5.6   | 5.6   | 5.4   | 5.4   | 5.2   | 4.7   | 4.8   | 4.8   | 4.6   | 4.6   | 5.0   | 5.9  |
| Precipitation                                                                    | PrecAvg     | 113.0  | 7.4   | 5.5   | 4.2   | 4.6   | 7.1   | 9.9   | 8.0   | 7.8   | 9.3   | 14.2  | 16.3  | 18.8 |
|                                                                                  | PrecMax     | 150.0  | 19.0  | 15.4  | 8.5   | 12.8  | 12.9  | 36.4  | 20.3  | 19.1  | 22.0  | 31.9  | 35.3  | 40.6 |
|                                                                                  | PrecMin     | 60.6   | 1.5   | 1.1   | 0.9   | 0.5   | 1.3   | 3.3   | 2.4   | 2.9   | 3.0   | 5.8   | 8.7   | 2.1  |
|                                                                                  | PrecStd     | 26.2   | 4.7   | 4.3   | 2.2   | 3.3   | 3.8   | 8.1   | 4.4   | 4.1   | 5.0   | 7.8   | 6.6   | 11.3 |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 86.1  | 86.8  | 87.8  | 89.1  | 91.7  | 92.3  | 91.5  | 92.5  | 91.8  | 90.0  | 88.9  | 87.8 |
|                                                                                  |             | MCWB   | 79.4  | 79.8  | 80.2  | 81.8  | 84.3  | 84.1  | 83.9  | 83.9  | 84.1  | 82.9  | 82.1  | 81.6 |
|                                                                                  | 2%          | DB     | 84.9  | 85.5  | 86.7  | 88.2  | 90.0  | 90.6  | 90.0  | 90.9  | 90.4  | 88.6  | 87.5  | 86.4 |
|                                                                                  |             | MCWB   | 78.7  | 79.0  | 79.8  | 81.5  | 83.2  | 83.5  | 83.1  | 83.4  | 83.6  | 82.3  | 81.4  | 80.8 |
|                                                                                  | 5%          | DB     | 84.0  | 84.8  | 86.0  | 87.7  | 89.2  | 89.7  | 88.9  | 89.7  | 89.2  | 87.6  | 86.7  | 85.4 |
|                                                                                  |             | MCWB   | 78.2  | 78.5  | 79.5  | 81.2  | 82.8  | 83.2  | 82.6  | 82.9  | 83.0  | 81.8  | 81.1  | 80.2 |
|                                                                                  | 10%         | DB     | 82.9  | 83.9  | 85.2  | 86.9  | 88.4  | 88.7  | 87.8  | 88.5  | 88.0  | 86.5  | 85.7  | 84.2 |
|                                                                                  |             | MCWB   | 77.5  | 77.9  | 79.0  | 80.8  | 82.4  | 82.7  | 82.1  | 82.3  | 82.4  | 81.3  | 80.7  | 79.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 81.3  | 82.1  | 82.8  | 84.7  | 86.3  | 86.7  | 85.9  | 86.6  | 86.4  | 84.5  | 83.9  | 83.4 |
|                                                                                  |             | MCDB   | 84.5  | 85.2  | 86.0  | 87.5  | 89.7  | 89.8  | 89.3  | 90.0  | 89.7  | 88.2  | 87.0  | 86.3 |
|                                                                                  | 2%          | WB     | 80.0  | 80.5  | 81.4  | 83.1  | 84.9  | 84.9  | 84.2  | 84.9  | 84.6  | 83.2  | 82.7  | 82.0 |
|                                                                                  |             | MCDB   | 83.6  | 84.3  | 85.4  | 87.0  | 88.7  | 89.1  | 88.5  | 89.4  | 88.8  | 87.4  | 86.4  | 85.3 |
|                                                                                  | 5%          | WB     | 79.0  | 79.4  | 80.3  | 82.0  | 83.7  | 83.8  | 83.2  | 83.7  | 83.5  | 82.4  | 81.8  | 80.9 |
|                                                                                  |             | MCDB   | 82.8  | 83.4  | 84.8  | 86.5  | 88.1  | 88.6  | 87.8  | 88.6  | 88.1  | 86.8  | 85.7  | 84.4 |
|                                                                                  | 10%         | WB     | 78.1  | 78.4  | 79.5  | 81.2  | 82.8  | 82.9  | 82.3  | 82.6  | 82.6  | 81.5  | 80.9  | 79.9 |
|                                                                                  |             | MCDB   | 81.8  | 82.6  | 84.0  | 86.0  | 87.5  | 87.8  | 87.2  | 87.7  | 87.4  | 85.8  | 84.9  | 83.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.0   | 9.4   | 10.3  | 11.2  | 11.0  | 10.4  | 9.9   | 10.3  | 10.5  | 9.8   | 9.2   | 7.9  |
|                                                                                  |             | MCDBR  | 9.7   | 10.9  | 11.6  | 11.9  | 12.0  | 12.0  | 11.9  | 12.1  | 12.6  | 11.6  | 10.8  | 9.9  |
|                                                                                  |             | MCWBR  | 5.7   | 6.5   | 6.8   | 7.3   | 7.3   | 7.3   | 7.0   | 7.0   | 7.5   | 7.2   | 6.5   | 6.2  |
|                                                                                  | 5% WB       | MCDBR  | 8.9   | 9.9   | 10.9  | 11.5  | 11.6  | 11.5  | 11.4  | 11.4  | 11.9  | 11.0  | 10.4  | 9.4  |
|                                                                                  |             | MCWBR  | 6.0   | 6.8   | 7.1   | 8.0   | 8.0   | 7.7   | 7.7   | 7.7   | 7.9   | 7.4   | 7.1   | 6.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.464  | 0.440 | 0.446 | 0.430 | 0.435 | 0.433 | 0.436 | 0.463 | 0.466 | 0.456 | 0.462 | 0.481 |      |
|                                                                                  | taud        | 2.282  | 2.368 | 2.355 | 2.414 | 2.427 | 2.446 | 2.439 | 2.342 | 2.326 | 2.355 | 2.316 | 2.224 |      |
|                                                                                  | Ebn at Noon | 265    | 277   | 277   | 279   | 273   | 271   | 271   | 267   | 268   | 269   | 263   | 255   |      |
|                                                                                  | Edh at Noon | 42     | 40    | 41    | 38    | 37    | 36    | 36    | 41    | 42    | 40    | 40    | 43    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1213   | 1534  | 1707  | 1952  | 1902  | 1741  | 1547  | 1607  | 1587  | 1436  | 1334  | 1142  |      |
|                                                                                  | RadStd      | 142    | 182   | 182   | 127   | 148   | 118   | 143   | 89    | 141   | 88    | 152   | 127   |      |

## Historical Trends

|      | DBAvg                  | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (5 neighbors) | +0.32   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page